

MODULE 3

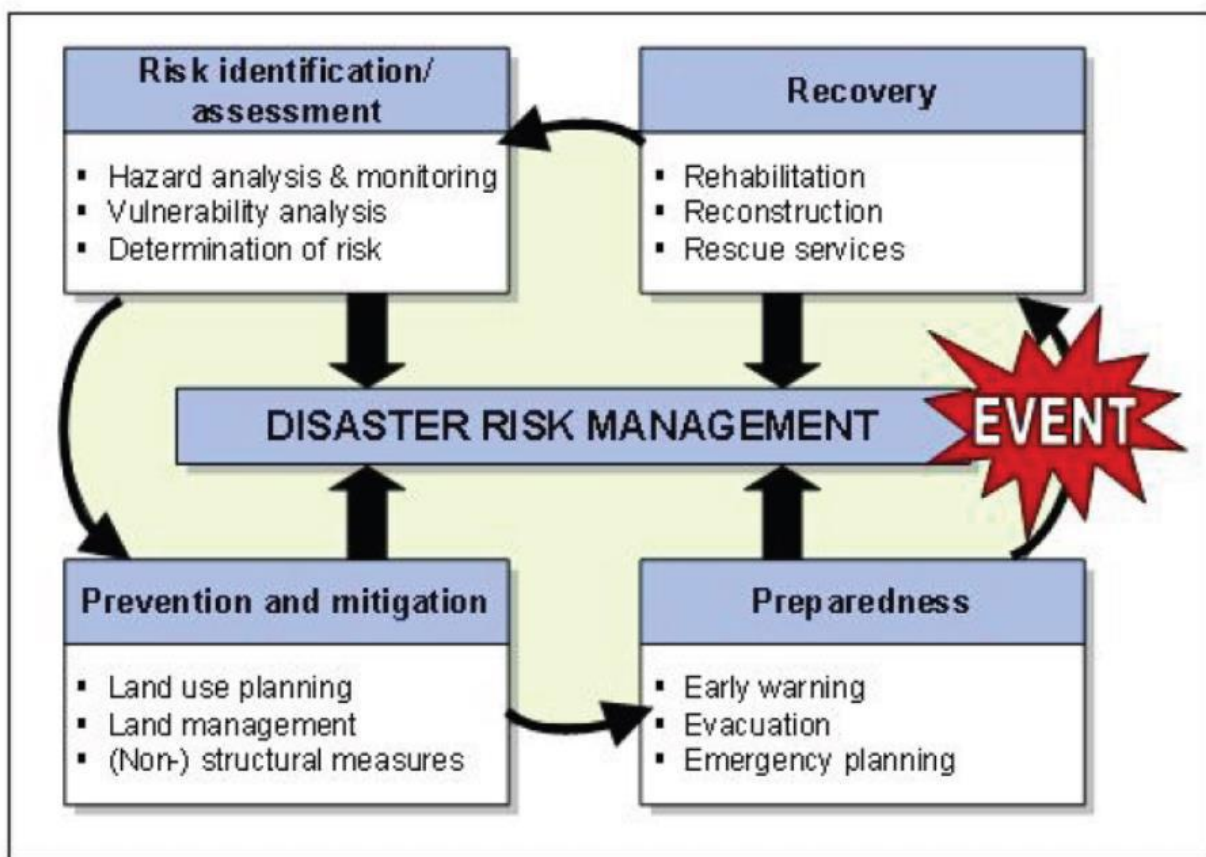
Disaster Risk Management

Definition of Disaster Risk Management (DRM)

The systematic process of using administrative directives, organizations, and operational skills and capacities to implement strategies, policies and improved coping capacities in order to lessen the adverse impacts of hazards and the possibility of disaster. Various frameworks have been proposed to understand the phases and process of disaster risk management.

Disaster Risk Management Framework

In this framework, the disaster risk management process (cycle) comprises the following main elements:



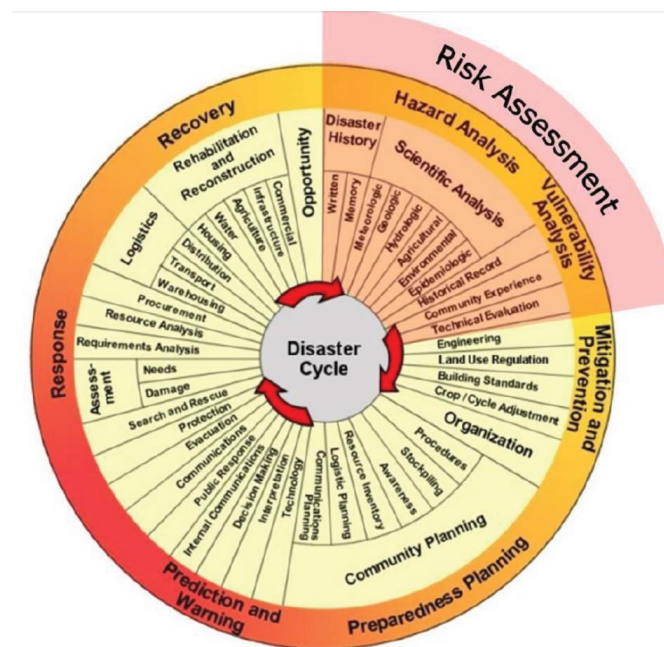
1. *Risk identification and assessment*: This involves determining and analyzing the potential, origin, characteristics and behavior of the hazard.

2. *Application of risk reduction measures in mitigation:* Planning and implementation of structural interventions or nonstructural measures such as disaster legislation.

3. *Disaster preparedness and emergency management:* Activities and measures taken in advance to ensure effective response to the impact of a hazard, including measures related to timely and effective warnings as well as evacuation and emergency planning.

4. *Recovery/ Reconstruction Decisions* and actions taken in the post-disaster phase with a view to restoring the living conditions of the affected population.

This Disaster Risk Management framework has several alternatives. In one of the alternatives, is that the disaster cycle is at the core and encompasses the usual elements of disaster risk management. Each of the core elements has been expanded to show most of the factors that are not easily discernible at the macro-scale.



Disaster Risk Reduction (DRR)

The concept and practice of reducing disaster risks through systematic efforts to analyse and manage the causal factors of disasters, including through reduced exposure to hazards, lessened vulnerability of people and property, wise management of land and the environment, and improved preparedness for adverse events.

DRR is limited in scope compared with DRM, which combines prevention, mitigation and preparedness with response. Given the importance of DRR in the international policy arena, five priority areas are underscored in the Hyogo Framework for Action (2005-2015):

1. Governance
2. Risk identification, assessment, monitoring and early warning
3. Knowledge management and education
4. Reducing underlying risk factors
5. Preparedness for effective response and recovery

The Hyogo framework for action was succeeded by the Sendai Framework for Disaster Risk Reduction 2015-2030 at the Third United Nations World Conference on DRR which took place in March 2015 in Sendai, Japan.

The four cornerstones of Disaster Risk Reduction

Four parallel and complementary lines of actions can be considered to reduce exposure to disasters and achieve a more sustainable approach to development:

1. Community / stakeholder participation
2. Public policy actions
3. Safer construction and urban development
4. Development of a culture of prevention

Disaster Prevention

Disaster Prevention is defined as those activities taken to prevent a natural phenomenon or potential hazard from having harmful effects on either people or economic assets. It refers to measures taken to eliminate the root causes that make people vulnerable to disaster. For disaster prevention to be successful, *a priori* planning is required. Planning of prevention hinges on two issues:

1. Hazard identification (identifying the actual threats facing a community) and
2. Vulnerability assessment (evaluating the risk and capacity of a community to handle the consequences of the disaster).

Once these issues are put in order of priority, emergency managers can determine the appropriate prevention strategies.

- Natural hazards cannot be prevented
- Human-made hazards can be prevented.
- Prevention has more to do with prohibiting man-made hazards.

However, mitigation and preparedness are considered as key measures of risk reduction for natural hazards.

Types of Disaster Prevention

Two types of disaster prevention includes:

1. **Primary prevention** is to reduce, avert or avoid the risk of the event occurring, by getting rid of the hazard or vulnerability, e.g. to avoid overcrowding, deforestation, choked drainage and to provide services.
2. **Secondary prevention** means to recognize promptly the event and to reduce its effects, e.g. by staying alert to possible displacements of population; by being ready to provide immunization, food, clean water, sanitation and health care to the affected population.

Disaster Preparedness

Disaster preparedness encompasses the knowledge and capacities developed by governments, professional response and recovery organisations, communities and individuals to effectively anticipate, respond to, and recover from, the impacts of likely, imminent or current hazard events or conditions

Components of a Comprehensive Disaster Preparedness Strategy includes:

- Hazard, risk and vulnerability assessments
- Response mechanisms and strategies
- Preparedness plans
- Coordination
- Information management
- Early warning systems
- Resource mobilisation
- Public education, training & rehearsals
- Community-based disaster preparedness

Types of Disaster Preparedness

1. Target-Oriented Preparedness: Preparedness plans may be target specific, for instance, we may require different types of planning for the vulnerable groups of women, children, elderly and disabled.
2. Task-Oriented Preparedness: Specific groups jointly develop activities based on one of the community's plans to evaluate the community's capability to activate the preparedness plan in a real emergency. Eventually, these tasks enable the development of plan revisions, employee training and material resources to support readiness.
3. Disaster-Oriented Preparedness: This addresses the likelihood of occurrence of a specific disaster. Emphasis is placed on structural and non-structural mechanisms.

Various levels of disaster preparedness in case of flooding is shown below:

Triggers	Early warning monitoring indicators	Responsible for early warning	Preventive and mitigating measures
Heavy rains	1) Weather forecasts 2) Information flow on the rainy season 3) Observation of the rise of river level	1) Gmet 2) Ministry of Information 3) NADMO	1) Use of media 2) Desilting of drainage systems 3) Clearing of waterways 4) Timely evacuation of population 5) Timely information flow on the rainy season
Opening of dams	1) Alert warning from Burkina Faso on the opening of dams 2) Rise of the water level in Burkina Faso	1) Volta River Authority 2) Ministry of Information 3) Ministry of Foreign Affairs 4) Ministry of Interior 5) NADMO 6) District Municipal and Metropolitan Assemblies	1) Timely flow of information on the opening of dams 2) Use of media 3) Use of voluntary groups by NADMO and red Cross 4) Evacuation plans and identification plans and identification of safer places
Choked drainage systems	1) Increment in human activities due to population increase resulting in unplanned settlements 2) No existing waste disposal sites	1) EPA 2) District and Metropolitan Assemblies 3) Ministry of Works and Housing	1) Establishment of an authority in charge of waste management 2) Clean up of drainage systems by District Assemblies

Importance of disaster preparedness

Disaster preparedness provides a platform to design effective, realistic and coordinated planning, reduces duplication of efforts and increase the overall effectiveness of government, household and community member's disaster preparedness efforts.

Disaster preparedness activities embedded with risk reduction measures can prevent disaster situations and also result in saving lives and livelihoods during any disaster situation, enabling the affected population to get back to normalcy within a short time period.

Disaster preparedness is a continuous and integrated process resulting from a wide range of risk reduction activities and resources rather than from a distinct sectoral activity by itself.

Disaster Response

Disaster responses are the set of activities taken during a disaster or immediately following a disaster, directed towards saving life and protecting property. The activities that deal with the effect of disaster may include medical care, evacuation, search and rescue, provision of emergency water, food and shelter, debris removal and stabilization of unsafe buildings and landforms.

The Objectives of Disaster Response

- It is aimed at providing immediate assistance to maintain life ,improve health and support the morale of the affected population.
- It is focused at meeting the basic needs of the people until more permanent and sustainable solutions can be found.
- It depends on the adequacy of preparedness.
- The success in responding appropriately depends on early planning, organization and training.
- Disaster response preparedness are the pre-disaster activities that are undertaken to minimise loss of life, injury and property damage in a disaster, and to ensure that rescue, relief, rehabilitation and other services can be provided following a disaster.
- Preparedness for the first and immediate response is referred to as“emergency preparedness”.

Factors that Determine the Nature of Disaster Response

1. *The type of disaster:* Disaster manifests in many forms, its onset may provide long warning, short warning, or no warning at all. It would influence the effectiveness of activating preparedness plans, mobilization, and application or response effort.
2. *The ability to take pre-impact actions:* Responses to disaster are operationalized in three main phases namely the “pre-during and post disaster ”situation. Disaster early warning systems may provide timely warnings for anticipating impending disaster. Pre-impact responses may be carried out if time and conditions are favourable.
3. *The severity and magnitude of disaster:* The severity and size of the problem determines the response required. Particular effects could be seen in the ability of responses to cope with the problem.
4. *The capability of sustained operations:* This is influenced by factors such as resource capability, management capability, community self-reliance, and availability of international assistance. It is important that these issues are clearly addressed in preparedness planning and response action plans.
5. *Identification of likely response requirements:* It is to be identified in the preparedness planning stage. The characteristics of disaster events might result in some or all of the following:
 - Many urgent and significant tasks involving injuries ,deaths ,or property loss/damage.
 - Large numbers of personnel and agencies required.
 - High levels of public interest and/or controversy.
 - Many examples of information mishandling.
 - Potential for some key tasks to be overlooked ,under-resourced ,over-resourced.
 - Potential for some of the available key resources to not be used.

Requirement for Effective Response

1. *Information:* An early warning system provides vital information for effective response operation. It must be robust to transmit warnings as early as practicable. Information gained

from these systems could help in the planning and decision-making as well as inform the general public.

2. *Resources:* It form an essential component of disaster response. The ability to mobilize the needed resources on short notice is most often hampered by many factors. The response plan as a component of the disaster management plan includes ways of managing human and financial resources, response to supplies availability and communication procedures. This involves identifying, strengthening ,and organizing resources and capacities for timely and effective response to a potential disaster.

Disaster Response Planning

In disaster response planning, roles and responsibilities are defined, policies and procedures are developed and generic tools for responses are identified and developed. Plans must be monitored, evaluated and adapted to the specific situation in times of disaster.

Rationale for Disaster Response

Responses are mainly directed at:

- limiting casualties;
- alleviating hardship and suffering;
- restoring essential life support and community systems;
- mitigating further damage and loss; and
- providing the foundation for subsequent recovery.

The Humanitarian Charter

The Humanitarian Charter provided four principles that must be followed in responding to emergencies based on the right to live in dignity, the right to receive humanitarian assistance, and the right to protection and security (The Sphere Project, 2011).

The principles are:

- avoid exposing people to further harm as a result of your actions;
- ensure people have access to impartial assistance;
- protect people from physical and psychological harm due to violence and coercion; and
- enable access to remedies and recovery from abuse (Sphere Handbook, 2012).

The Sphere Project aims to improve the quality of assistance provided to people affected by disasters, and to improve the accountability of humanitarian actors to their constituents, donors and affected people. This Project has suggested four minimum standards that often need consideration during response to disasters.

These are:

1. Water supply, sanitation and hygiene promotion;
2. Food security and nutrition; and
3. Shelter, settlement and non-food items; and
4. Health action.

Disaster Responses

Disaster responses include actions that embrace the following:

- Search and rescue
- First aid and emergency medical care
- Evacuation
- Evacuation centre management
- Development of Standard Operation Procedure (SOPs)
- Immediate repair of community facilities and services
- Relief delivery
- Coordination and Communication
- Psycho-social counselling and stress debriefing
- Medical services

Search and rescue

This activity is usually conducted by well-trained volunteers in finding disaster victims, that is, lost, sick or injured persons in either a remote or difficult to access areas such as water bodies, desert, forest or probably in the course of mass population movement. They are often directed at, locating endangered persons at an emergency incident, removing those persons from danger, treating the injured, and providing for transport to an appropriate health care facility.

Basic Key Steps of Search and Rescue

1. Sizeup - involves assessing the situation and determining what one is going to do and if yes, then how.

The decision whether to attempt a rescue should be based on:

- a. The risks involved ;and
- b. Achievement of the overall goal of doing the greatest good for the greatest number.

2. Search-involves locating victims and documenting their location.

3. Rescue-involves the procedures and methods of extricating
And moving victims to safety.

•First aid and emergency medical care

First aid is the provision of initial care for an illness or injury.

It is usually performed by a nonexpert, but trained personnel to a sick or injured person until definitive medical treatment can be accessed.

Emergency medical care is immediate paramedic attention to severe wounds and the rapid transportation of the ill or injured to a health facility.

•Evacuation

Evacuation is an organized movement of people from an area at risk to a safer place.

Types of Evacuation

Precautionary evacuation before disaster

Protective evacuation after disaster

Evacuations for reconstruction purposes

Services provided during evacuations include:

Registration

Assistance with financial and legal queries

Water, food, clothing

Rest areas

Blankets and personal items

Interpreter services

Assistance in contacting family/friends

Services for animals

Emergency financial assistance

First aid, medical and health

Information

- Standard Operating Procedures (SOPs)

SOPs are the standard procedures that “operationalize” the disaster response and/or contingency/plans. The SOPs set out what should be done, how it should be done, who is responsible for implementing what, and specifies available resources.

- SOPs take cognizance off our stages of preparation and procedures:

during normal times

alert/warning and

during disaster

rehabilitation

During Normal Times

The state institution mandated to respond to disaster ought to:

1. Formulate and distribute disaster preparedness plans, and conduct drills in all areas;
2. Produce maps of Wards/Village Tracts showing areas most vulnerable to storms, floods and other natural disasters;
3. Make a list of vehicles and motor boats that can be used for emergency work;
4. Compile a list of departments, non-governmental organisations (NGOs), and members of People’s Strength that will take part in relief operations in the predisaster, disaster and post-disaster periods and designate representatives for contact;
5. Obtain beforehand the required relief and aid supplies;
6. Form the necessary disaster preparedness committees and organizations;
7. Create shelters and safe locations for use during disasters depending on local conditions;
8. Conduct educational talks on natural disasters and rehearse periodically for the local community depending on local conditions; and Coordinate with departments concerned to

form Security services, Auxiliary Fire Brigades, communication agencies and Red Cross Societies; and

9. Provide organising and training activities.

Alert/Warning Stage

1. Emphasize the dissemination of news obtained through early warning systems to the community;
2. Assign duties to administrative bodies and NGOs to fly warning flags as part of the disaster preparedness programme in the vulnerable areas of the Ward/Village Tract;
3. Alert and mobilize members of the Security services, Auxiliary Fire Brigade, communication agencies, the Red Cross, Youth, members of People's Strength and NGOs;
4. Make the necessary arrangements to evacuate the public to safe locations (shelters) in a timely manner;
5. Increase security sentries as required;
6. Ensure that all levels of supervisors have all teams ready for assigned duties; and
7. Keep the office operational 24 hours a day in the emergency period.

During Disaster Stage

1. Alert the community in areas the natural disaster is likely to strike;
2. Safeguard the road and water transport routes, keep relief and medical teams at the ready and arrange transport to affected areas at short notice;
3. Evacuate the community from vulnerable areas to safe locations or designated shelters as quickly as possible;
4. Operate relief camps and supervisory centers at designated shelters as quickly as possible;
5. Ensure that administrative personnel and NGOs in areas vulnerable to storms give disaster warnings door to door as a matter of urgency;
6. Keep available relief and aid supplies at the ready to launch relief operations quickly and effectively;
7. Evacuate the public remaining in the area to designated safe locations;

8. Make arrangements to evacuate movable property including cattle to designated locations; and
9. Ensure the well-disciplined implementation of orders received from the coordinating agencies and sub-committees with the help of members of the Security services, Fire Brigade, Red Cross Youth members, and members of People's Strength, social organisations and NGOs.

Rehabilitation Stage

The SOPs for the rehabilitation stage are:

1. Conduct field inspections in affected areas as soon as possible and provide the necessary assistance and support;
2. Submit immediate preliminary reports with population figures, death and injury figures of cattle and animals, data on socio-economic losses, and carry out further systematic data collection;
3. Make arrangements to provide health care and social protection to disaster victims;
4. Clear collapsed buildings and trees as quickly as possible;
5. Prioritise the restoration of transportation, electricity and water supply and telephone and telegraph services as soon as possible;
6. Make arrangements as quickly as possible to reclaim contaminated wells and ponds for access to clean water and dig new wells for drinking water;
7. Make arrangements to bury/cremate the remains of disaster casualties and animal carcasses;
8. Manage and systematically utilize disaster funds and supplies, as well as cash and supplies donated by well-wishers, social organizations and NGOs; and
9. Support the local population for the resumption and recovery of economic activities to previous conditions.

• Relief Aid

This relates to any provision of assistance during an emergency that is meant to attend to a person's immediate requirements for survival or recovery.

It may include food, clothing, housing, medical care, necessary social services and security when a person is faced with circumstances beyond her or his control. Relief aid must be targeted at the most vulnerable first: Vulnerable children or orphans, female or child headed households, pregnant or lactating women, sick or elderly populations.

•Coordination and Communication

Meetings -Plan to hold regular meetings with Movement partners to determine activities and roles;

Information management -Information sharing on disaster impact, assessment and needs through input into DMIS (Disaster Management Information System);

Communication means -Plan for continuous information and communication flow; Movement Coordination Frame works -Plan for strategic and operational coordination; and

Partnership agreements -Identify existing agreements and determine additional agreements needed to meet needs.

Accurate and comprehensive information is often a requisite without which response operations would be difficult. The setting of Emergency Operations Centres(EOC) is essential for the effective management of information. EOCs ensure that information is correctly processed according to the proven cycle of:

acquisition of information;

information assessment;

decision-making; and

dissemination of decisions and information.

• Psychosocial Support

Impacts on psychosocial well-being can be both short term and long term.

Psychological services play a crucial role in responding to crises that involve large populations.

They help in the recovery process and reduce the development of mental health problems.

Psychosocial support activities include identifying and referring individuals requiring specialized support through professional mental health services.

- **Public Health Services**

The relevance of medical services is most felt in when there are:

- Deaths, injuries
- Loss of clean water, shelter, sanitation, routine hygiene
- Disruption of solid waste management
- Public concern for safety
- Increased pests and vectors
- Damage to health care system
- Worsening of chronic illnesses
- Toxic/hazardous exposure
- Loss of food supply
- Standing surface water

The public health services required in responding to disasters include:

- Mass casualty management
- Mental health
- Environmental health
- Reproductive health
- Managing and continuation of existing health services
- Managing and continuation of medication on chronically affected diseases (HIV, TB, Leprosy, etc.)
- Management of the dead and missing
- Emergency feeding
- Communicable disease surveillance and response
- Sanitation

Relief

- **Relief** refers to the provision of essential, appropriate and timely humanitarian assistance to those affected by a disaster.
- It is defined as the provision of assistance or intervention during or immediately after a disaster to meet the life preservation and basic subsistence needs of those people affected.
- Relief can be of an immediate, short term, or protracted duration.
- Relief, as a disaster management process, provides timely essential needs such as basic household items, shelter, food, water and sanitation, or health items.

• Relief activities

provide goods and services to disaster-affected populations in the form of supplies, vouchers or cash transfers, so as to enable those populations to cover their essential needs.

Relief measures differ, depending upon the nature of disaster. At certain occasions, money may have no value, but certain articles like food, clothes, etc. may be more important.

The disasters that require relief may include fire, flood, lightening, earthquake landslides and sinking of land, snow avalanches, glaciers, drought, locusts, epidemics, land storms, excessive rains, cloud bursts and hailstorm.

Other issues that may also require relief are accidents like motor vehicles accident, boat mishap, food poisoning and fall from tree or steep rock, Non-explosive burst and snakebite.

Relief, globally, is guided by fundamental principles which disaster organisations and NGOs engaging in disaster relief response ought to apply.

The principles require that:

- Response to disasters must have humanitarian imperative
- Aid is provided based on needs alone and must be done without discrimination of any kind.
- Aid will not be used to further a particular political or religious standpoint
- Aid agencies do not to act as instruments of government foreign policy
- That culture and custom are respected in response and relief activities

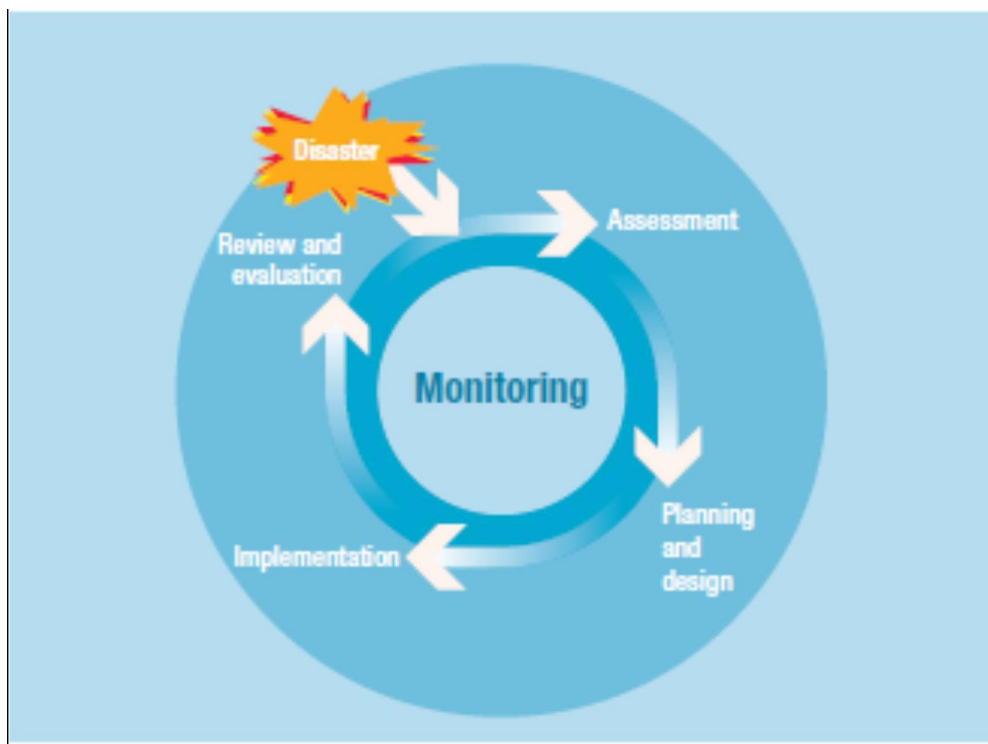
- Disaster response is built on local capacities
- Ways be are developed to involve programme beneficiaries in the management of relief aid
- Relief aid would reduce future vulnerabilities to disaster as well as meeting basic needs
- Accountable to both those we seek to assist and those from who we accept resources
- Information, publicity and advertising activities shall recognize disaster victims as dignified human beings and not hopeless objects.

•Project Cycle Management Linked to Relief

The project cycle management (PCM) is a conceptual tool used for the planning and management of programmes and projects leading to the improvement of programme effectiveness and outcome.

In disaster response, the PCM provides five main continuous components that guide disaster response.

They are: assessment, planning and designing, implementation, monitoring, review and evaluation



•Assessment

The first step after the onset of a disaster is to assess the origin, magnitude and effect of the disaster on the affected population so as to be able to identify humanitarian needs and plan possible interventions.

The purpose is to ascertain the need for an intervention based on identified needs.

There are four types of assessments: immediate, rapid, detailed and continual assessments.

•Assessment

Immediate assessments are conducted within 72 hours after the disaster. It is generally conducted by in country act or sand involves the collection of basic disaster information.

Rapid assessments are conducted immediately after the disaster and take upto a week. These involve gathering of information on the needs and existing capacities of the affected population.

Detailed assessments are conducted fter rapid assessments to obtain further information on the affected population's needs and capacities for programme planning. They can take upto a month to conduct ,more or less depending on the area ,the complexity of the issues and resources available.

Continual assessments are conducted once the detailed assessment has been completed and relief programmes are running. They update information on the situation and involve beneficiaries for recovery programming.

•Planning/Design

The planning and design process is critical to a relief operation as it helps to set out in clear stages what the operation will do. It also determines how progress and results will be monitored and defines the duration of relief activities. The planning process describes the steps and order to undertake when planning relief interventions. The design describes the selection of relief interventions that are best adapted to the needs of the context faced.

The factors that must be considered when planning relief activities are:

The needs identified during the assessment, presented by technical sector

The overall plan of action for the operation

The capacity and mandate of the disaster management organisation

Programmes conducted/planned by other stakeholders

The security and access to disaster-affected populations

• **Implementation**

The implementation phase is when the activities are carried out to achieve the desired results. The implementation of relief activities is conducted once the assessment and the planning processes are completed. This may include the mobilisation and distribution activities. The key issues that would be considered are beneficiary targeting and identification and methods and tools for implementation.

Beneficiary Targeting

The relief intervention, depending on the nature and magnitude of the disaster and the capacity to respond, can aim to cover all the population or can be focused on the most vulnerable groups within the affected population. The purpose of targeting is to provide relief to those who are least able to cover their essential needs. In slow onset emergencies and/or when relief is provided more than once to a population, targeting may be necessary. In rapid onset emergencies, relief is often provided to all of the population in the affected location, given the lack of time to conduct proper targeting.

Beneficiary Identification

Beneficiary identification is conducted prior to or during any relief operation in any disaster condition although very difficult to conduct in rapid disaster situations. In rapid disaster situation the minimal information on the number of persons requiring supply is enough to start the distribution process. However, in slow onset disasters that may require more than one relief operation, a detailed identification process would be required. Beneficiary identification serves two main purposes. Firstly, it helps to identify persons in need; secondly, it helps in ensuring accountability and judicious use of resources.

Generally, the identification process requires information such as:

head of household's given name and surname

number of family members

address or location of permanent and/or temporary residence

In situations that require long-term programmes where multiple distributions are planned, it may be advisable to collect supplementary information for family members such as: name, gender, age, national identification number, date of identification and any other relevant information such as pregnant mothers, physically challenged etc.

- **Monitoring**

Monitoring, also known as process evaluation, is a continuous process of follow-up on all the activities throughout the disaster response and relief activities. Monitoring is done through the use of indicators which measure the extent to which an objective has been accomplished. These indicators measure changes in the lives of beneficiaries as a result of the relief they received. The resulting progress reports lead to decisions on whether or not an intervention needs to be changed or adapted as the situation evolves.

- **Review and Evaluation**

Reviews and evaluations are the assessments carried out at a given point in the intervention. These involve analysis of the situation that produces intermediate/final reports that lead to recommendations for and adaptations to the intervention or to lessons learnt for the organisation. Reporting is essential for proper internal and external communications, transparency and accountability. Examples of key reports done during the life-cycle of the project are assessment narrative reports and relief activity reports.

- **Institutions for Disaster Relief and Response**

Many organisations respond to disasters both locally and internationally. In the local front, the leading government agency in disaster response is the National Disaster Management Organisation. The security agencies, such as the Police, Armed Forces, Navy, Air Force and the National Fire Service are also key in disaster response in the country. Other governmental organisation ssuch as the National Ambulance services provide health emergency services.

Non-governmental organizations (NGOs) and civil society groups operating within the area of impact may also respond depending on their mandate and capacity.

• List of International Relief Organisations

The list of international organizations providing response and relief are enormous.

The list of international relief response organizations is as follows:

Action Against Hunger (AAH), CARE, Caritas Internationalis, Catholic Relief Services, (CRS-USCC), Emergency Nutrition Network (ENN), Doctors Without Borders, Food For The Hungry International (FHI), FoodForTheHungry, HungerPlus, Inc., Interaction, International Committee of the Red Cross (ICRC), International Federation of Red Cross and Red Crescent Societies (IFRC), International Organisation for Migration (IOM), International Rescue Committee (IRC),.....Lutheran World Federation, Mennonite Central Committee (MCC), Mercy Corps (MC), Overseas Development Institute (ODI), Oxfam, Refugees International, Relief International, Save the Children, The Office of U.S. Foreign Disaster Assistance (OFDA), United Nations Children's Fund (UNICEF), United Nations High Commissioner for Refugees (UNHCR), United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA), US Committee for Refugees (USCR), World Vision International.

Disaster Mitigation

Disaster mitigation refers to the lessening or limitation of the adverse impacts of hazards and related disasters. The adverse impacts of hazards often cannot be prevented fully, but their scale or severity can be substantially lessened by various strategies and actions. Mitigation measures encompass engineering techniques and hazard-resistant construction as well as improved environmental policies and public awareness.

Primary Objectives of Disaster Mitigation

The primary objectives of disaster mitigation are two fold,

1. Hazard likelihood reduction:

Appropriate for a few natural hazards, as it is not possible to reduce the occurrence of many hazards. However, the likelihood of floods occurrence can be reduced by mitigation measures such as sea defence walls.

2. Risk consequence reduction:

This is a reduction in the impact of a hazard, via a reduction in exposure and/or vulnerability.

It involves ensuring that the population, structures, or other systems are able to withstand such an event with as few negative consequences as possible.

The primary aim in both is to decrease risk of death and injury to the population and the secondary aims are to decrease damage and economic losses inflicted on public sector infrastructure and to reduce private sector losses.

Types of Disaster Mitigation Measures

- Structural Mitigation Measures: Any physical construction to reduce or avoid possible impacts of hazards, which includes engineering measures and construction of hazard resistant and protective structures and infrastructure.
- Non-structural Mitigation Measures: This refers to policies, awareness, knowledge development, public commitment, and methods and operating practices, including participatory mechanisms and the provision of information, which can reduce risk and related impacts.
- Structural and non-structural disaster mitigation measures are mainly carried out by human beings.
- However, nature through ecosystem functions also provides several mitigation measures as shown in next slide.

Hazard and disaster mitigation functions of ecosystems

Ecosystem	Hazard Mitigation
Mountain forests, vegetation on hillsides	Vegetation cover and root structures protect against erosion and increase slope stability by binding soil together, preventing landslides. Catchment forests, especially primary forests, reduce risk of floods by increasing infiltration of rainfall and delaying peak floodwater flows, except when soils are fully Saturated. Forests on watersheds are important for water recharge and purification, drought mitigation and safeguarding drinking water supply.

Wetlands, floodplains	Wetlands and floodplains control floods in coastal areas. Marshes, lakes and floodplains release wet season flows slowly during drought periods.
Coastal ecosystems (mangroves, Salt marshes, coral reefs, sand dunes)	Coastal wetlands, tidal flats, deltas and estuaries reduce the height and speed of storm surges and tidal waves. Coastal ecosystems protect against storm surges, flooding and other coastal hazards – combined protection by coral reefs, sea grass beds and sand dunes/ coastal wetlands/coastal forests is particularly effective.
Drylands	Natural vegetation management and restoration in drylands contributes to ameliorate the effects of drought and control desertification, as trees, grasses and shrubs conserve soil and retain moisture. Prescribed burning and creation of physical firebreaks in dry landscapes reduce fuel loads and the risk of unwanted large-scale fires.